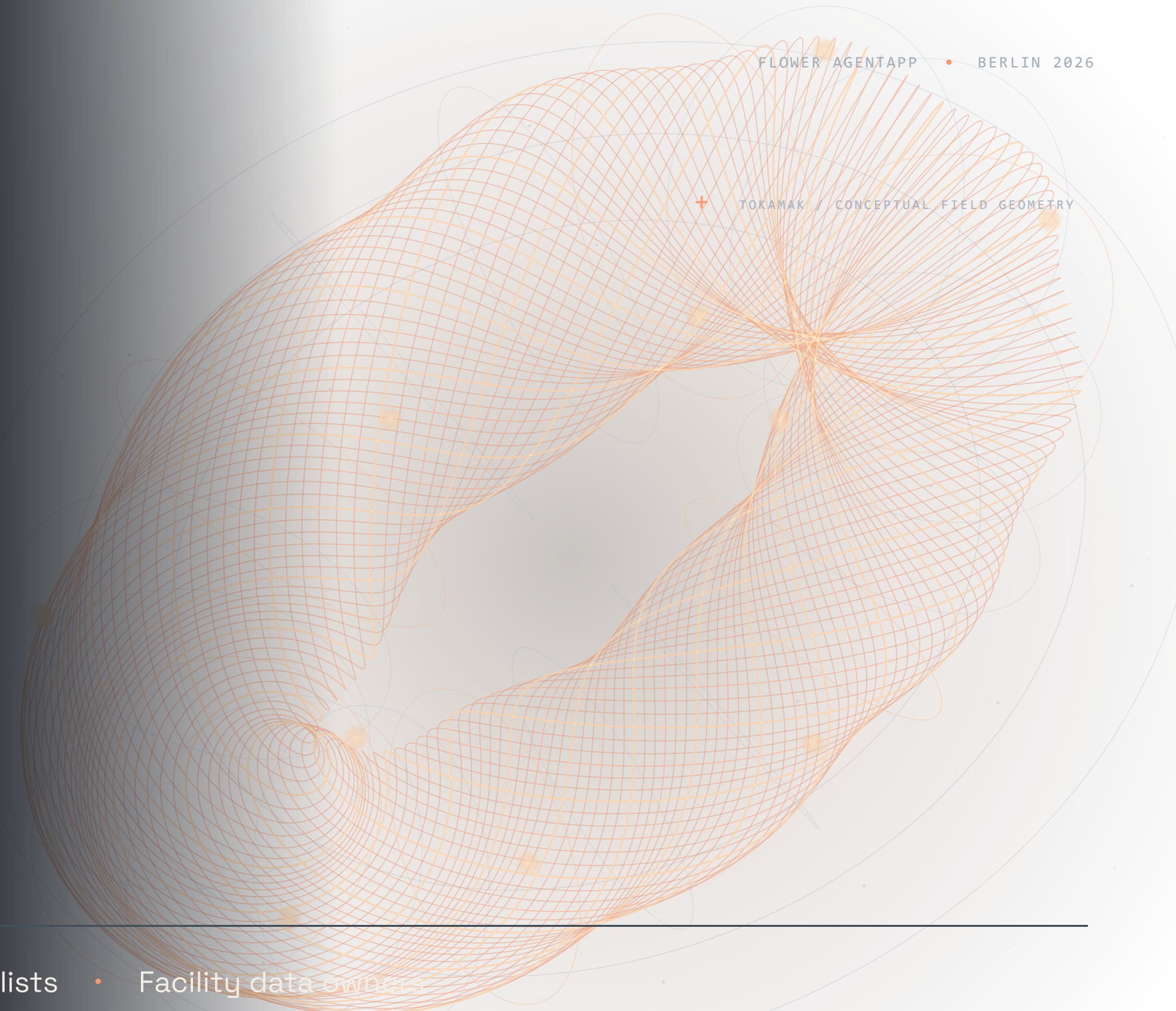


— 01 / WHAT IT IS • WHO IT SERVES

Investigate across sites. Keep control local.

A Flower AgentApp for evidence-led
fusion research between experiments.

BUILT FOR Experimental physicists • Diagnostic specialists • Facility data owners



02 / WHAT GOES IN • WHAT COMES OUT

A question in. A next experiment out.

RESEARCHER → INVESTIGATOR

“Why did facility A respond weakly to heating?”

+ configured site tools

01 / LOCAL ANALYSIS CONSUMES

Scientific inputs

- + Device parameters
- + Temperature and density profiles
- + Heating-source profiles

Synthetic profiles • reduced 1-D solver.

▫ Underlying arrays stay in local analysis.

02 / THE MODEL RECEIVES

Approved evidence

- + Site capabilities + machine context
- + Transport band + 11 profile samples
- + Source-audit findings or refusals

Each request passes the disclosure gateway.

➔ Fixed products, with evidence identifiers.

03 / THE RESEARCHER RECEIVES

A testable next step

- + Report with evidence citations
- + Uncertainty and missing evidence
- + Recommended next measurement

Tool results retained alongside the report.

✂ Inspect releases, refusals and provenance.

EXAMPLE OUTPUT

“Independently verify delivered heating at A.”

03 / WHY THIS SOLUTION FITS

Flexible investigation. Firm disclosure rules.

APPROACH	FIT FOR THIS WORKFLOW
Pool raw data	Useful when pooling is permitted. Requires authority to export the source data.
Prompt-only agent	Flexible reasoning. Instructions alone do not enforce disclosure.
Fixed workflow	Repeatable, explicit request sequence. A strong baseline for recurring investigations.
Our harness ↗	Evidence-driven request choice. Code-enforced releases and a traceable record.

Best fit: share findings under site-controlled disclosure.

Research benefit over a fixed workflow remains to be measured.

THE ENFORCEABLE PART

1 → 2 → 2

Context → balance → source check

5 units

Disclosure allowance per site and case

-  Cached releases.
Spending persists with the ledger.
-  Applicability checks.
Steward approval cannot override policy.

Flower supplies the runtime. We supply the science.



FLOWER AGENT HARNESS

Infrastructure we can reuse.

- 01 AgentApp execution + model endpoint
Run the application and route its model calls.
- 02 Streamed events + run-series state
Present reports and retain app-defined evidence.
- 03 Local and hosted execution
Develop with SuperLink; submit runs to SuperGrid.

FUSION INVESTIGATOR

Where our work belongs.

Scientific tools

Local transport analysis and structured findings

Bounded investigation

Tool limits, steward contexts and report instructions

Disclosure gateway

Python rules and a site-controlled SQLite ledger

Build the investigation on an existing runtime.

[Flower runtime documentation](#) ↗

Default sites share one process; independent site services are supported.

Follow the evidence.

EVIDENCE SOURCE

Recorded gateway • no LLM
Captured Python outputs • works offline

A Facility A
DIII-D-like • synthetic
CAUSE UNRESOLVED

No independent power audit.
Next step: measure delivered heat.

5 / 5 units spent

B Facility B
SPARC-like • synthetic
AUDIT AVAILABLE

Corrected: near reference.
Delivered power below assumed.

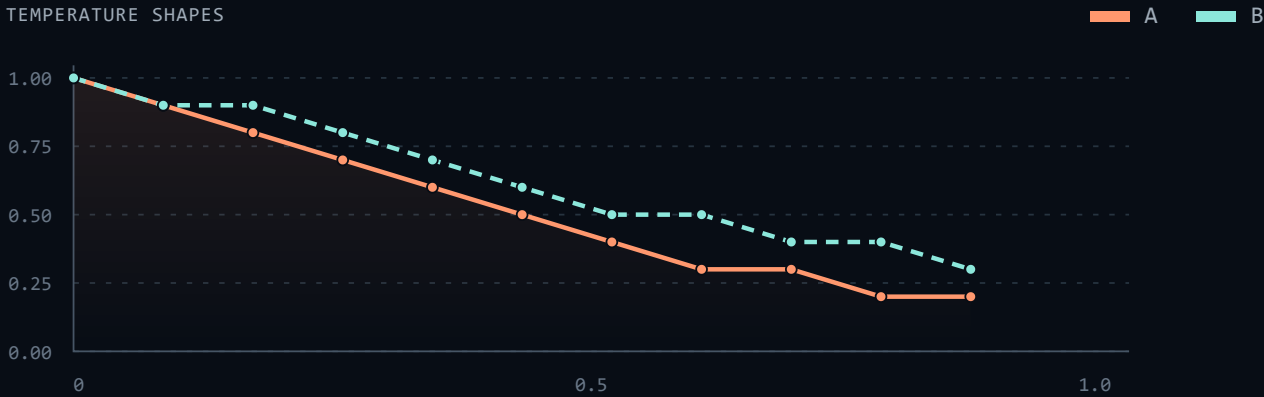
5 / 5 units spent

C Facility C
TCV-like • synthetic
BALANCE DENIED

Analogy not applicable.
No balance profile released.

1 / 5 units spent

RELEASED TEMPERATURE SHAPES



11 rounded samples • normalized radius r/a • synthetic reduced 1-D model

CACHED RELEASE

Same finding. No second charge.

Facility B returns its previously released source evidence.
Spending remains 5 / 5.

EVIDENCE 178c623af9a14240 • RECORDED reduced-1d-transport-v1

01 A • Context	02 A • Balance	03 B • Context	04 C • Context	05 C • Balance	06 B • Balance	07 B • Source	08 A • Source	09 B • Raw logs	10 B • Source
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06 / THE RESEARCHER'S NEXT STEP

Measure the power that arrived.

Independently verify delivered heating at facility A.

A Unresolved
Needs an independent source audit.

B A useful precedent
Audited transport returns near reference.

C Analogy refused
Fails the demo's gradient criterion.

THE VALUE OF THE
HARNESS

A better next question.
A traceable path to it.

